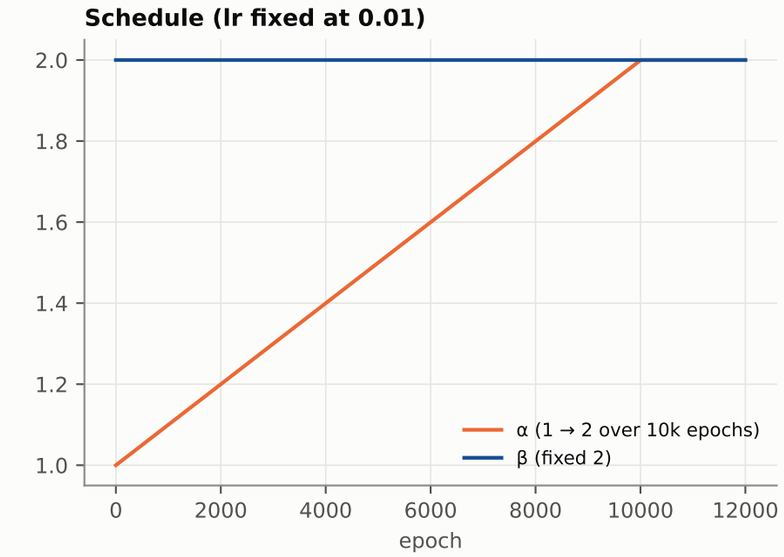
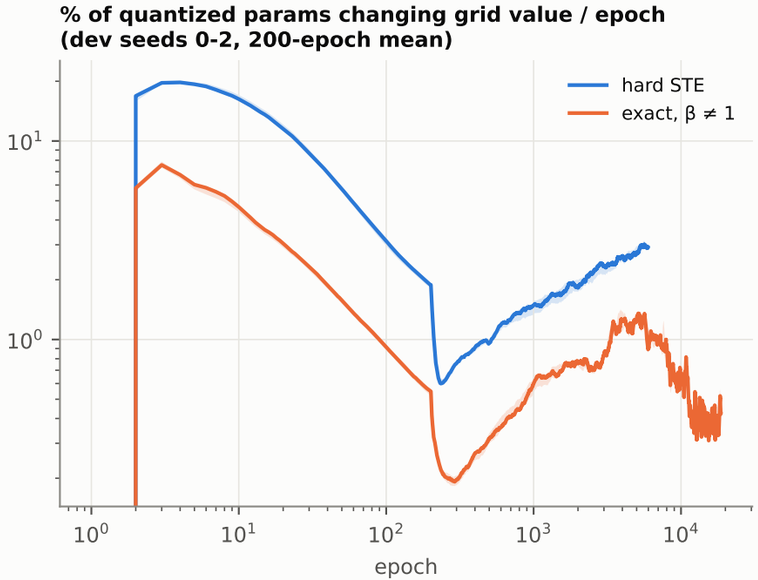
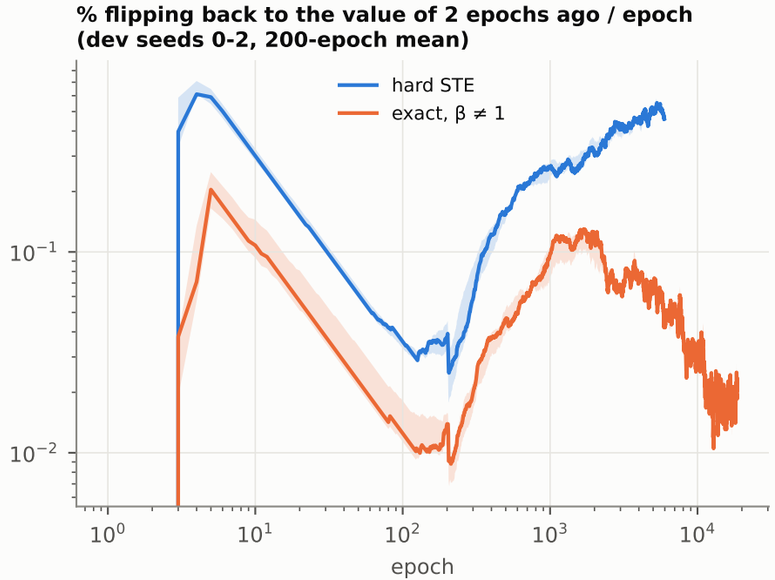
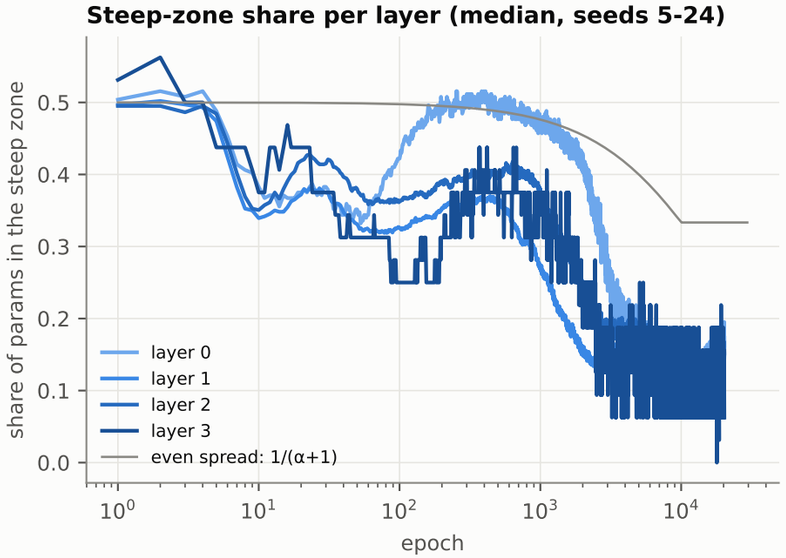
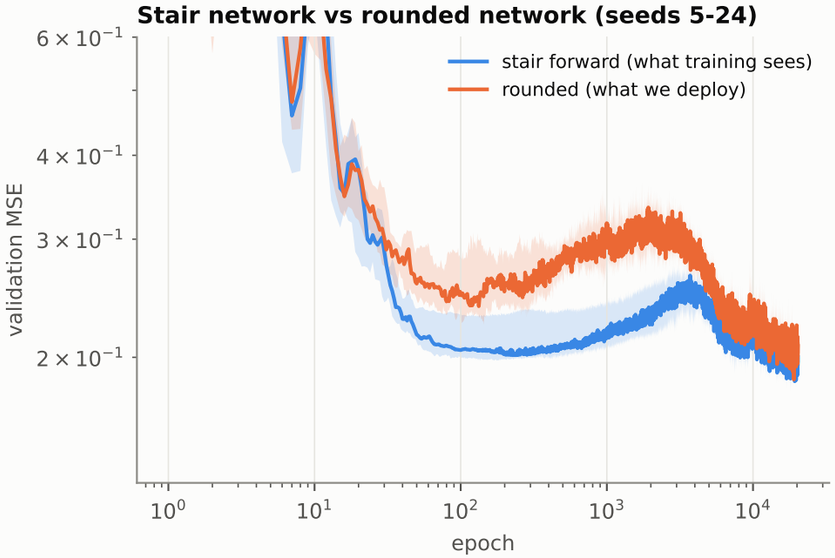
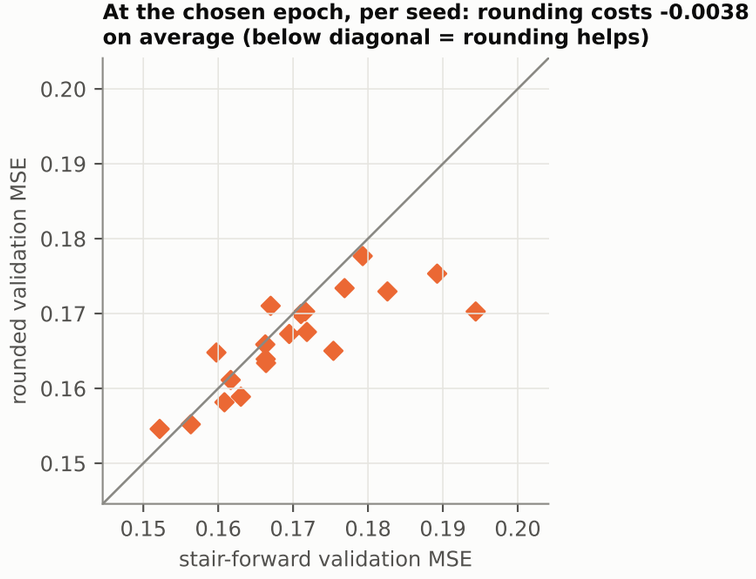

Schedule (lr fixed at 0.01)
α (1 → 2 over 10k epochs)
β (fixed 2)
epoch


% of quantized params changing grid value / epoch
(dev seeds 0-2, 200-epoch mean)
hard STE
exact, β ≠ 1
epoch


% flipping back to the value of 2 epochs ago / epoch
(dev seeds 0-2, 200-epoch mean)
hard STE
exact, β ≠ 1
epoch


Steep-zone share per layer (median, seeds 5-24)
share of params in the steep zone
layer 0
layer 1
layer 2
layer 3
even spread: 1/(α+1)
epoch


Stair network vs rounded network (seeds 5-24)
validation MSE
stair forward (what training sees)
rounded (what we deploy)
epoch


At the chosen epoch, per seed: rounding costs -0.0038
on average (below diagonal = rounding helps)
rounded validation MSE
stair-forward validation MSE